

① Dynamical systems, described by diff. equations

Ex: Velocity control of a car

$$\dot{y}(t) = G_1 y(t) + G_2 u(t) + G_1 d(t)$$

where output ($y(t)$) : what we want to control
→ velocity

input ($u(t)$) : what we use to control it
→ pedal

disturbance ($d(t)$) : what we cannot control
→ wind

How we can solve it? ② Using Laplace transform

$$\mathcal{L}\{y(t)\} = \int_0^{\infty} y(t) \cdot e^{-ts} \cdot dt = Y(s)$$

Important transforms =

step response

$$2.1 \quad u(t) = \begin{cases} 0 & t < 0 \\ A & t \geq 0 \end{cases} \Rightarrow \mathcal{L}(u(t)) = \int_0^{\infty} A \cdot e^{-st} dt = -\frac{A e^{-st}}{s} \Big|_0^{\infty} = \frac{A}{s}$$

$$2.2 \quad \dot{u}(t) \Rightarrow \int \frac{du}{dt} \cdot e^{-st} dt \Rightarrow \int du v = v \cdot u - \int u dv$$

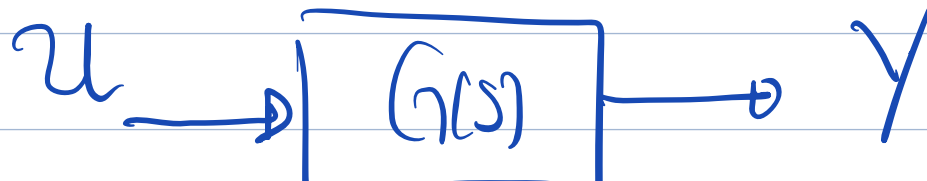
$$= u \cdot e^{-st} \Big|_0^{\infty} + s \int_0^{\infty} u \cdot e^{-st} dt = -u(0) + s \cdot U(s)$$

$\mathcal{L}(\dot{u}(t)) = U(s)$

$$\mathcal{L}(\dot{u}(t)) = sU(s) - u(0)$$

$$2.3 \quad \int_0^t u(\tau) d\tau = \frac{1}{s} U(s) \quad (\text{do it yourself})$$

③ Transfer function



$$G(s) = \frac{\mathcal{L}(y(t))}{\mathcal{L}(u(t))} = \frac{Y(s)}{U(s)}$$

Example

$$\dot{y}(t) = -a y(t) + u(t) \quad \text{for } y(0)=0$$

$$sY(s) - y(0) = -aY(s) + U(s)$$

$$Y(s)(s+a) = U(s) \Rightarrow \frac{Y(s)}{U(s)} = \frac{1}{(s+a)}$$

no poles

we call it

$$\frac{P(s)}{Q(s)} = \frac{Y(s)}{U(s)}$$

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↳ zeros

Poles and zeros

Poles =

- Poles of the system are roots of the denominator polynomial $D(s)$

- Poles that are closer to the imaginary axis result in slower responses*, and complex poles with larger imaginary parts result in higher frequency of oscillations.

→ * (dominant poles = poles that significantly shape the system's response)

→ $(\tau s + 1) \rightarrow \tau$: time constant

EX: $\frac{1}{(s+1)(s+2)} \Rightarrow$ poles $s = -1$ and $s = -2$

WTS/021

dominant pole is -2 (slow

$= \frac{1}{10(0.2s+1)(0.5s+1)} \Rightarrow$ ^{THE 4% RULE:} 4% to reach about 98% of its final value

• Poles determine stability of the system:

Poles in the RHP $\rightarrow$ unstable system

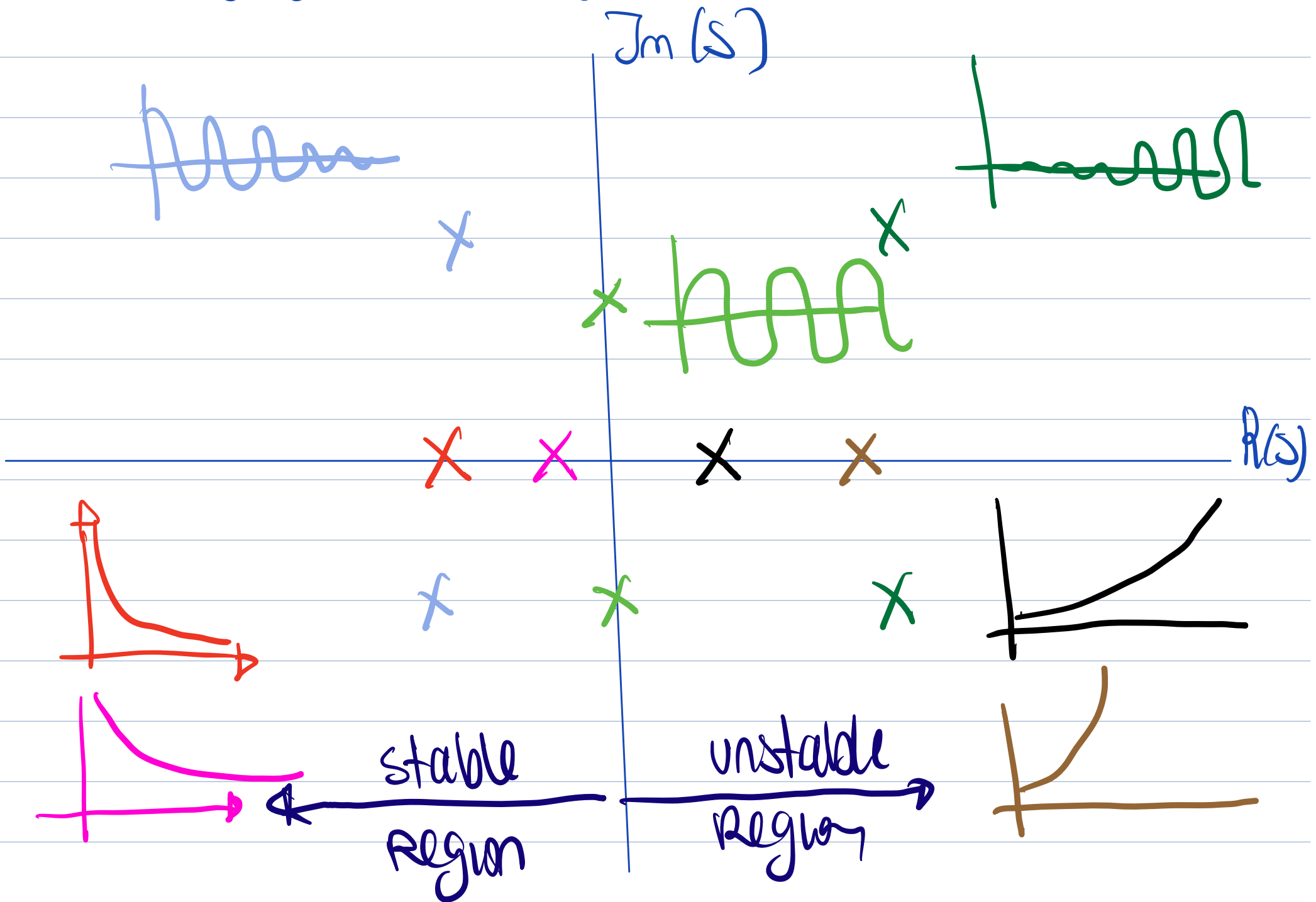
" " " LHP $\rightarrow$ stable

Zeros:

• Zeros of the system are the roots of the numerator, $P(s)$

• Do NOT influence system stability, only transients

• Can induce overshoot



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Final value theorem (FVT) ^{output}

If all the poles of $sY(s)$ are in the LHP of the s -plane, then

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$$

6 Is the value for which the response of the system converges after all transients have died out. To find the DC gain, we assume that there is a unit step input $[U(s) = 1/s]$ and we use the FVT

$$\text{DC gain} = \lim_{s \rightarrow 0} s(G(s)) \frac{1}{s} = \lim_{s \rightarrow 0} G(s) = G(0)$$

2.8 Consider a system described by the model $Y(s) = G(s)U(s)$ as shown in Figure 2.8a. Given $G(s)$, how is the step response computed? How can the step response be determined using experiments?

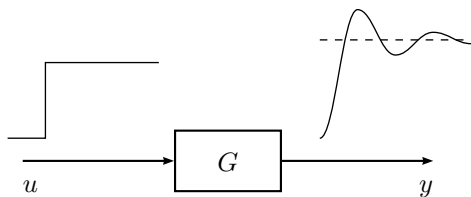


Figure 2.8a

2.9 Figure 2.9a shows the step response of a system $Y(s) = G(s)U(s)$. The input step has amplitude 1. Use the figure and determine

- Steady state value.
- Overshoot M in % of the final value.
- Rise time T_r .
- Settling time T_s .

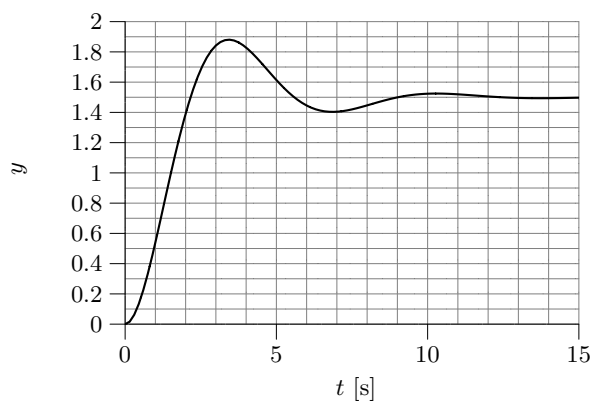


Figure 2.9a

2.10 Figure 2.10a shows the step responses of four different systems. Combine each step response with a transfer function from the alternatives below.

	Transfer function	Poles	Zeros	$ G(0) $
C	$G_1(s) = \frac{100}{s^2 + 2s + 100}$	$-1 \pm 10i$		1
	$G_2(s) = \frac{1}{s+2}$	-2		1/2 X
B	$G_3(s) = \frac{10s^2 + 200s + 2000}{(s+10)(s^2 + 10s + 100)}$	$-10, -5 \pm 8.7i$	$-10 \pm 10i$	2
	$G_4(s) = \frac{200}{(s^2 + 10s + 100)(s+2)}$	$-2, -5 \pm 8.7i$		1
	$G_5(s) = \frac{600}{(s^2 + 10s + 100)(s+3)}$	$-3, -5 \pm 8.7i$		2
	$G_6(s) = \frac{400}{(s^2 - 10s + 100)(s+2)}$	$-2, 5 \pm 8.7i$		2

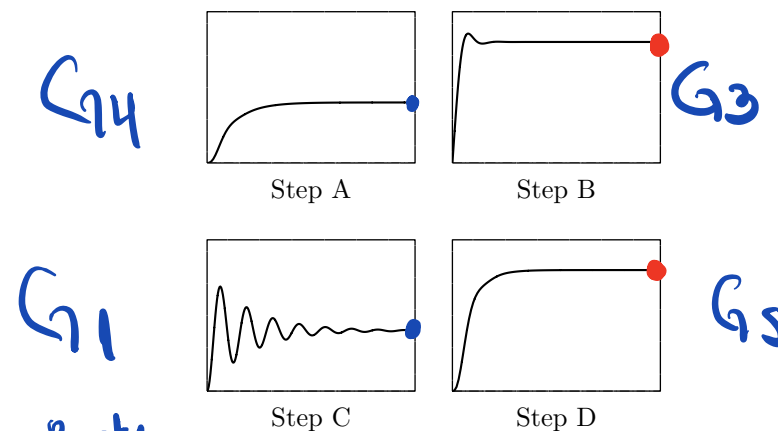


Figure 2.10a. All comparable axes have equal scaling.

- $|G_A(0)| = |G_C(0)|$ & $|G_B(0)| = |G_D(0)|$
 $G_2(s)$ ~~X~~
- Unstable pole at $s=5$
 $G_6(s)$ ~~X~~
- $|G(0)|=1$ which is slower? $G_1(s)$
 $\rightarrow G_4(s)$ dominant pole is -2

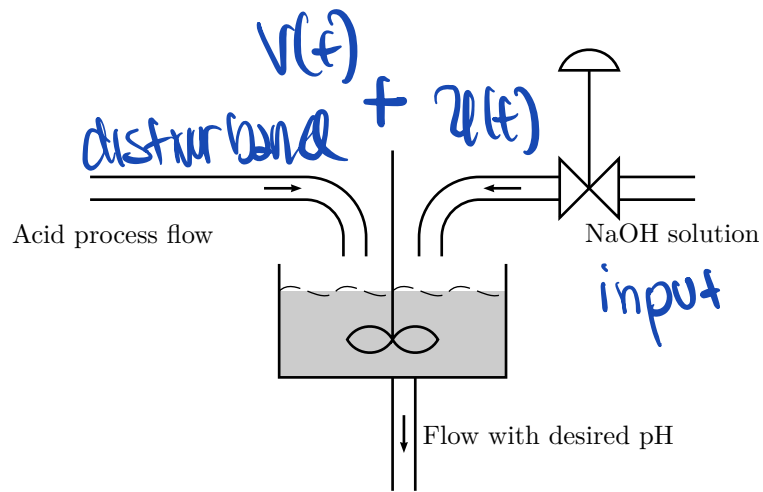
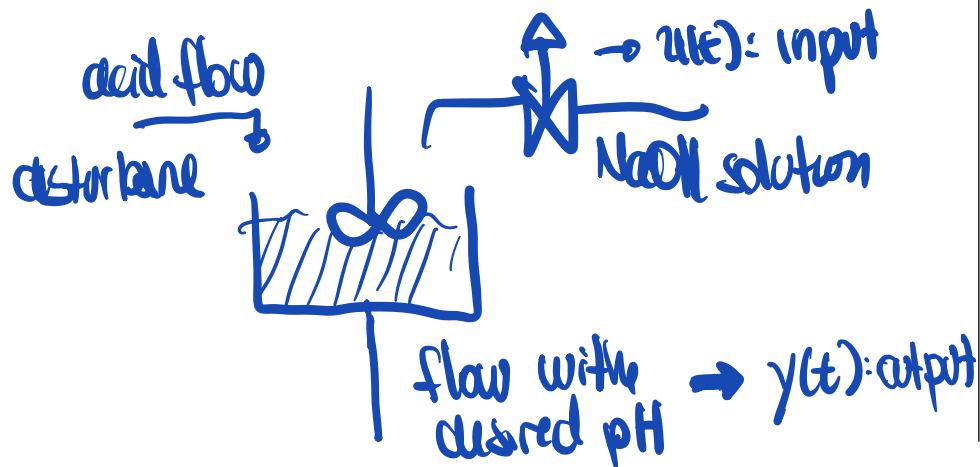


Figure 2.11a

2.11 In the continuously stirred tank, see Figure 2.11a, an acid process flow is neutralized by adding a concentrated NaOH solution. The acid process flow has a tendency to vary its pH with time, which gives undesired variation of the pH in the outflow. In an effort to reduce these variations one has decided to use control.

- Classify the different signals as input, output, and disturbance signal.
- Draw a block diagram of the system with a control strategy.



so the oscillatory behavior is not dominating the step response
 ④ zero: overshoot - which is slower? (s/s)

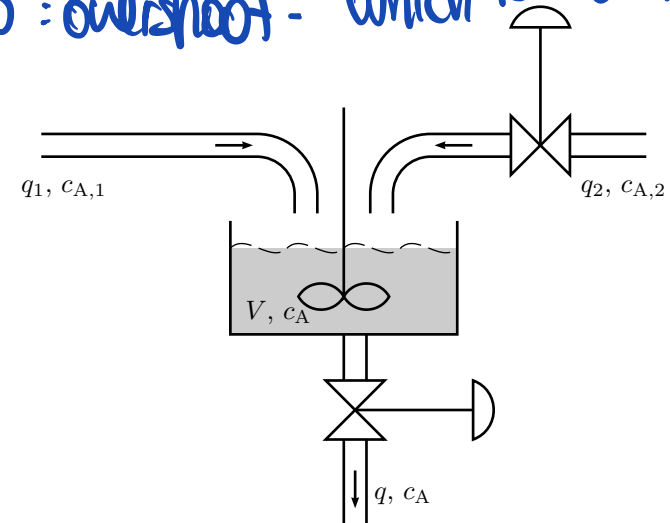


Figure 2.12a

2.12 Two flows with different concentrations of a chemical component A are mixed in a continuously stirred tank, see Figure 2.12a. The volume can be assumed constant, $V = 1 \text{ m}^3$.

- During a stationary period the values of the concentrations and flows are, $c_{A,1}^* = 1.0 \text{ kmol/m}^3$, $q_1^* = 1.0 \text{ m}^3/\text{min}$, $c_{A,2}^* = 4.0 \text{ kmol/m}^3$ and $q_2^* = 0.5 \text{ m}^3/\text{min}$. What are $c_A(t)$ and $q(t)$ during this period?
- Write down the dynamical balance equation for component A. State all your assumptions. Is the dynamical model linear?
- Assume that $c_{A,1}$ changes value from 1.0 kmol/m^3 to 1.2 kmol/m^3 at $t = 0$. Show that the expression for $c_A(t)$ can be written as

$$c_A(t) = k_0 + k_1 (1 - e^{-t/\tau})$$

Determine the constants k_0 , k_1 and τ .

2.11,

